

PROJECT REPORT

End-to-End Data Pipeline, AI Clustering
and Interactive Dashboard

HOLIDAY ITINERARY

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Introduction

1.1 Context

The tourism industry generates massive amounts of geospatial and behavioral data every day. Modern travelers often face an overwhelming number of choices when planning trips, especially in large cities containing hundreds or thousands of points of interest (POIs). This abundance of information frequently leads to inefficient itinerary planning, unnecessary travel time, and poor overall travel experiences.

At the same time, advances in Data Engineering, Machine Learning, and Geospatial Analytics now make it possible to build intelligent systems capable of transforming raw tourism data into optimized travel recommendations.

This project addresses this challenge by designing and implementing a complete end-to-end data platform capable of:

- Collecting and processing tourism-related datasets.
- Structuring data using modern Medallion Architecture principles.
- Combining relational and graph databases.
- Applying Machine Learning clustering techniques.
- Generating optimized multi-day itineraries.
- Delivering recommendations through an interactive dashboard.

The final system demonstrates how modern data infrastructure can be used to create intelligent, scalable, and user-friendly tourism recommendation services.

1.2 Research Question

The main research question of this project is:

How can a robust data engineering pipeline integrate relational and graph databases to generate intelligent, location-optimized travel itineraries using Machine Learning techniques?

This central question raises several important technical sub-problems:

- How can heterogeneous tourism datasets be ingested and cleaned efficiently?
- How can geospatial information be modeled for fast querying?
- What database technologies are best suited for structured and relationship-oriented tourism data?
- How can Machine Learning optimize trip organization geographically?
- How can the generated itineraries be exposed through APIs and dashboards?
- How can the entire pipeline remain scalable and reproducible?

1.3 Project Objectives

1.3.1 Data Engineering Objectives

- Build a complete ETL pipeline following the Medallion Architecture.
- Ingest and clean tourism datasets from raw sources.
- Store optimized datasets in Parquet format.
- Enrich geospatial information using H3 indexing.

1.3.2 Infrastructure Objectives

- Deploy all services using Docker containers.
- Combine PostgreSQL and Neo4j in a polyglot persistence architecture.
- Build scalable backend services using FastAPI.
- Ensure reproducibility and modularity.

1.3.3 Artificial Intelligence Objectives

- Apply unsupervised Machine Learning techniques.
- Use K-Means clustering for geographic itinerary optimization.
- Group POIs into optimized daily travel plans.

1.3.4 Frontend Objectives

- Build an interactive Streamlit dashboard.
- Visualize itinerary results dynamically.
- Provide real-time metrics and maps.

1.4 Overall Solution Overview

The proposed solution is a modular end-to-end architecture composed of five major layers:

1. Data Ingestion Layer
2. Data Processing and Transformation Layer
3. Storage Layer (SQL + Graph Databases)
4. API Layer

5. Visualization Layer

The system collects raw tourism datasets, processes them through a Medallion pipeline, enriches them with geospatial metadata, and applies clustering algorithms to generate optimized travel itineraries.

The entire solution is containerized using Docker and exposed through REST APIs and a Streamlit dashboard.

2

Overall Project Architecture

2.1 Overall Architecture

The architecture follows a modular and scalable workflow:

Data Sources → ETL Pipeline → PostgreSQL / Neo4j → FastAPI → Streamlit Dashboard

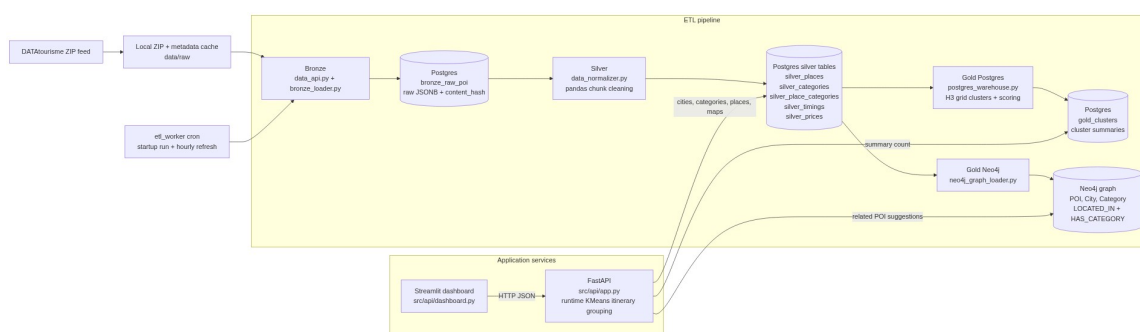


Figure 2.1: Overall architecture of the Holiday Itinerary platform

The system was designed following modern Data Engineering principles:

- Separation of concerns.
- Scalable storage.
- Reproducible infrastructure.
- Modular deployment.
- Real-time API exposure.

2.2 Component Description

2.2.1 Data Sources

The project uses tourism-related datasets containing:

- Points of Interest (POIs)

- Geographic coordinates
- Categories
- Ratings and metadata
- City information

Data sources are initially stored in CSV and JSON formats.

2.2.2 ETL Layer

The ETL pipeline transforms raw datasets into structured analytical datasets.

The ETL process performs:

- Data cleaning
- Missing value handling
- Type normalization
- Geospatial enrichment
- H3 indexing
- Clustering preparation

2.2.3 Storage Layer

The storage layer combines two complementary technologies:

- PostgreSQL for structured analytical storage.
- Neo4j for graph-based recommendation logic.

2.2.4 API Layer

FastAPI exposes REST endpoints used by the dashboard.

The API provides:

- Health monitoring endpoints
- Itinerary generation endpoints
- POI retrieval services
- Cluster visualization data

2.2.5 Visualization Layer

The Streamlit dashboard allows users to:

- Select cities
- Choose travel duration
- Filter by interests
- Generate optimized itineraries
- Visualize recommendations on maps

2.3 Technology Choices and Justification

2.3.1 PostgreSQL

PostgreSQL was selected as the main relational database because of:

- ACID compliance
- Excellent geospatial support
- Efficient analytical querying
- Reliability and scalability

The structured nature of tourism datasets makes PostgreSQL particularly suitable for storing cleaned POI data.

2.3.2 Neo4j

Neo4j was chosen to manage complex relationships between entities.

Graph databases are highly efficient for:

- Relationship traversal
- Recommendation systems
- Path optimization
- Similarity exploration

This allows future implementation of advanced recommendation engines.

2.3.3 FastAPI

FastAPI was selected due to:

- High performance
- Native asynchronous support
- Automatic Swagger documentation
- Pydantic-based validation

2.3.4 Streamlit

Streamlit enables rapid dashboard development entirely in Python.

Advantages include:

- Simple deployment
- Interactive widgets
- Native map visualization
- Fast prototyping

2.3.5 Docker

Docker ensures:

- Environment reproducibility
- Dependency isolation
- Simplified deployment
- Service orchestration

Data Collection and Engineering Pipeline

Data collection is the critical starting point of the entire production pipeline. Without reliable, up-to-date, and well-structured geospatial information, no Machine Learning clustering model or graph recommendation engine can produce accurate, logistically viable results for the end-user. This section describes how official French tourism market data is systematically retrieved from the national **DATA**tourisme platform, the two collection and synchronization modes implemented via background workers, and how this raw data is staged and validated before being processed by the downstream Medallion layers.

3.1 Data Source: The DATAtourisme Semantic Registry

The core data source for this ecosystem is the official **DATA**tourisme open-data platform. This national registry aggregates scattered regional, communal, and departmental tourism data streams across France into a unified index. The ingested dataset includes:

- Detailed Points of Interest (POIs) with precise geographic coordinates (Latitude and Longitude).
- Multi-level taxonomic categories (e.g., museums, historical monuments, restaurants, parks).
- Dynamic structural metadata including operational opening hours, seasonal schedules, and explicit pricing constraints.

This dataset is published as a deeply nested semantic graph packaged within a compressed multi-file ZIP configuration, which requires robust automated extraction pipelines to be converted into analytical relational formats.

Listing 3.1: Raw JSON-LD Source Data Schema

```
{
  "@id": "String (URI)",
  "@type": ["String"],
  "rdfs:label": {
    "fr": "String",
    "en": "String"
  },
  "geo": {
```

```
"latitude": "Decimal",
"longitude": "Decimal"
},
"isLocatedAt": {
  "postalCode": "String",
  "addressLocality": "String"
},
"lastUpdate": "ISO-Timestamp"
}
```

3.1.1 Implemented Ingestion Modes

To ensure optimal data freshness while protecting the platform against API rate limiting or network drops, the background orchestration layer (`etl_worker cron`) implements two distinct collection modes:

1. **Bootstrap/Startup Execution Mode:** When the Docker containers are initially provisioned and the microservices spin up, a startup script automatically triggers the `data_api.py` component. This script executes a complete download of the latest compressed national ZIP feed and populates the local disk cache directory (`data/raw`) to guarantee that the application has an immediate baseline dataset to work with.
2. **Incremental Synchronization Mode (Hourly Refresh):** Tourism data is dynamic; scheduling updates, pricing changes, or seasonal events occur frequently. To handle these changes without disrupting the user-facing web applications, an automated background cron job executes an hourly data refresh. This daemon checks for file deltas, updates the local `data/raw` cache, and feeds new data streams seamlessly into the pipeline.

3.2 Data Engineering Pipeline

3.2.1 The Medallion Architecture

The project follows the Medallion Architecture pattern, which organizes data into progressive refinement layers.

Listing 3.2: Holiday Itinerary Repository Layout

```
data/
  raw/
  silver/parquet/
src/
  pipeline.py
  bronze/
    bronze_loader.py
    data_api.py
  silver/
    data_normalizer.py
  gold/
    postgres_warehouse.py
    neo4j_graph_loader.py
```

3.2.2 Bronze Layer

Once the dataset is pulled by `data_api.py`, it is immediately handed over to the ****Bronze Layer**** pipeline managed by `bronze_loader.py`. At this landing stage, the data is still un-

parsed and raw. The pipeline processes it as follows:

- The raw records are written directly into a high-performance PostgreSQL table column named `bronze_raw_poi` utilizing the native binary JSONB format.
- To prevent the storage layer from duplicating identical rows during recurring hourly synchronization cycles, the system generates a unique cryptographic MD5 signature for each incoming record (`content_hash`). If a record's hash already exists in the staging environment, the row write is omitted, achieving absolute pipeline **Idempotency**.

The Bronze layer contains raw ingested datasets exactly as received from external sources.

Characteristics:

- Raw CSV and JSON files
- No transformations
- Immutable storage
- Historical preservation

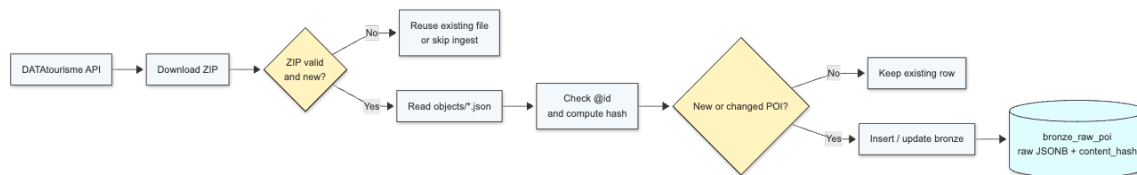


Figure 3.1: Data pipeline ingestion flow illustrating the transfer of raw JSON-LD tourism payloads from external APIs into the immutable PostgreSQL Bronze storage layer.

3.2.3 Silver Layer

The Silver layer contains cleaned and standardized datasets.

Transformations include:

- Data cleaning
- Null value management
- Schema normalization
- Duplicate removal
- Conversion to Parquet format

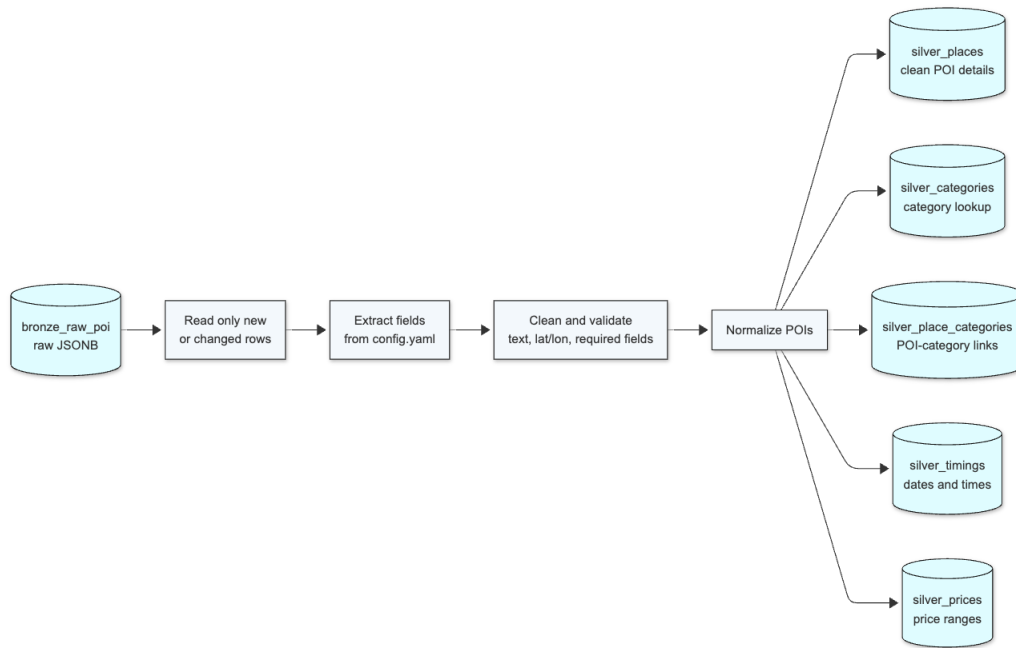


Figure 3.2: The data normalization pipeline within the Silver layer, transforming nested JSON-LD fields, standardizing geographic coordinates, and mapping entities into a relational PostgreSQL schema

Parquet format was chosen because it:

- Reduces storage size
- Improves query performance
- Supports columnar analytics
- Optimizes large-scale processing

3.2.4 Detailed Architectural Mechanics of the Silver Layer

The Silver Layer within the Medallion Architecture represents the transition from raw data collection to structural refinement. In this platform, the primary execution is driven by the `data_normalizer.py` engine, which subscribes to strict relational normalization principles.

Memory Optimization via Chunked Ingestion

Because the upstream national tourism feed from DATAtourisme encompasses hundreds of thousands of complex records, a monolithic data load would instantly induce Out-Of-Memory (OOM) errors within the isolated Docker containers. To secure a predictable and low memory footprint, the Silver parser leverages the native streaming functionality of the Pandas library:

Listing 3.3: Chunksize Memory Boundary Enforcement in Silver Layer

```

import pandas as pd

def process_bronze_to_silver(db_connection, chunk_size=10000):
    """Streams and transforms raw JSONB payloads in atomic memory
    boundaries."""
    query = "SELECT payload FROM bronze_raw_poi"
    for chunk_df in pd.read_sql_query(query, db_connection, chunksize=
        chunk_size):

```

```
# Execute structural transformation and text parsing
sequentially
normalize_and_load_chunk(chunk_df)
```

Data Cleaning, Deduplication, and Structural Normalization

The transformation pipeline executes a deterministic multi-stage cleaning routine upon every micro-batch stream:

- **JSON Flattening:** Deeply nested semantic JSON-LD structures are mapped and transformed into rigid tabular arrays.
- **Coordinate Strict Validation:** Remic geolocation fields are parsed into high-precision floating-point numbers. Any record containing corrupt, empty, or mathematically impossible Latitude/Longitude coordinates is purged to shield downstream Machine Learning models from distortion.
- **Relational Mapping and Normalization:** The unstructured data tree is partitioned across structured entity tables: `silver_places`, `silver_categories`, `silver_timings`, and `silver_prices`. This relational database state ensures that complex analytical queries do not require repetitive string-parsing overhead.

3.2.5 Gold Layer

The Gold layer contains enriched business-ready datasets.

Additional enrichments include:

- H3 geospatial indexing
- Cluster identifiers
- Precomputed analytics
- Aggregated statistics

This layer is directly consumed by the API and dashboard.

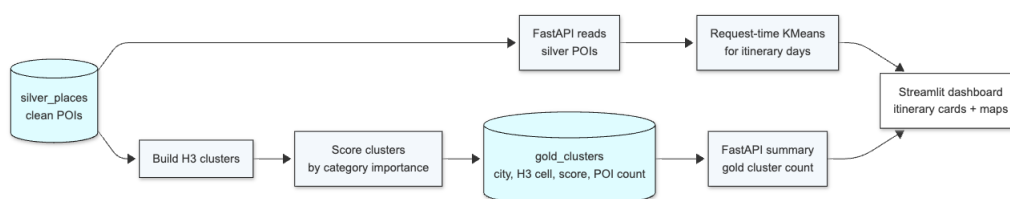


Figure 3.3: End-to-end analytical pipeline interface, showcasing how optimized data and graph metrics from the Gold tier are served directly to the interactive Streamlit user dashboard

3.2.6 Detailed Architectural Mechanics of the Gold Layer

The Gold Layer, often referenced as the Business or Presentation Layer, represents the pinnacle of data refinement within the Medallion Architecture. In this ecosystem, the Gold layer serves as the specialized analytical engine that directly powers the FastAPI microservices and the interactive Streamlit interface. Rather than executing heavy queries on structural tables at runtime, the Gold layer hosts pre-aggregated, indexed, and optimized business-ready views.

Geospatial Partitioning via Uber H3 Hexagonal Indexing

To facilitate ultra-low-latency spatial queries and proximity lookups, the Gold Postgres pipeline (`postgres_warehouse.py`) enriches the clean coordinates from the Silver layer using **Uber's H3 Spatial Indexing System**. Traditional administrative boundaries vary too widely in size to be useful for mathematical distribution analytics. The system projects every latitude and longitude into a uniform Resolution-8 hexagonal grid cell ID. This provides two massive engineering advantages:

- **Spatial Bucket Pre-computation:** Points of interest are pre-grouped into localized spatial buckets on the database level, drastically lowering the coordinate search space for the active K-Means machine learning engine.
- **Analytical Performance:** Statistical summaries (e.g., total landmark density per hex-bin, structural popularity indexes) are materialized into the `gold_clusters` summary table, allowing the front-end dashboard to surface real-time KPIs instantly.

Table 3.1: PostgreSQL Relational Database Schema and Data Types

Table	Column	Type	Purpose / Description
bronze_raw_poi	payload	JSONB	Raw nested JSON-LD from DATAtourisme.
	content_hash	VARCHAR(32)	MD5 hash for pipeline idempotency.
silver_places	place_id	UUID	Primary Key. Unique identifier.
	name	VARCHAR(255)	Sanitized title of the POI.
	city	VARCHAR(100)	Standardized municipality name.
	latitude	NUMERIC(9,6)	Cleaned latitude for K-Means.
	longitude	NUMERIC(9,6)	Cleaned longitude for K-Means.
silver_categories	category_id	INT	Primary Key. Unique category index.
	label	VARCHAR(100)	Category name for user preferences.
silver_prices	price_id	SERIAL	Primary Key. Auto-increment ID.
	min_price	NUMERIC(6,2)	Minimum entry ticket in Euros.
	max_price	NUMERIC(6,2)	Maximum entry ticket in Euros.
silver_timings	timing_id	SERIAL	Primary Key. Timing configuration index.
	open_time	TIME	Daily standard opening hour.
	close_time	TIME	Daily standard closing hour.
gold_clusters	h3_index	VARCHAR(15)	Primary Key. Uber H3 hexagonal ID.
	poi_count	INT	Pre-computed sum of locations in hex.
	density_score	NUMERIC(3,2)	Popularity score for rendering.

Topological Graph Modeling inside Neo4j

A prominent feature of this architecture's Gold layer is its decoupled **Polyglot Persistence Layer**. While PostgreSQL manages the structured analytics, the script `neo4j_graph_loader.py` maps relational states into an optimized network topology inside the Neo4j graph database.

Data is refactored from tables into semantic nodes and high-degree vector relationships:

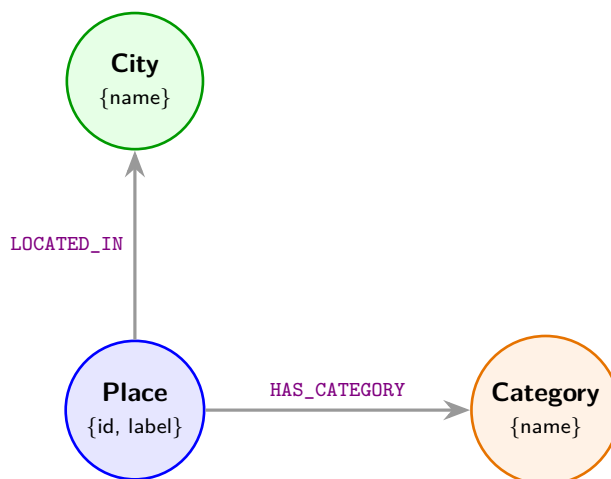


Figure 3.4: Neo4j Semantic Graph Database Schema

3.3 Infrastructure and Containerization

To guarantee local reproducibility, environment isolation, and seamless cross-platform deployment, the entire microservice architecture is fully containerized using **Docker**. Managing multiple polyglot storage engines alongside independent application layers introduces significant deployment complexity; therefore, **Docker Compose** is utilized as the core infrastructure orchestrator.

3.3.1 Service Topology and Orchestration

The infrastructure defines a decoupled network topology where services communicate through an isolated virtual network bridge. This setup prevents external dependency conflicts and secures database access boundaries. The application ecosystem is distributed across the following core architectural components:

- **PostgreSQL Instance:** Serves as the primary relational backend, maintaining the sequential states of the Medallion pipeline (Bronze raw JSONB, Silver cleaned tables, and Gold precomputed summary aggregates).
- **Neo4j Graph Infrastructure:** Operates as the native graph database engine, handling Cypher index queries for relationship-aware POI and city recommendations.
- **FastAPI Backend Gateway:** An asynchronous ASGI web server running within an optimized Python container, executing runtime machine learning clustering (K-Means) and exposing secure RESTful JSON interfaces. **Streamlit Presentation Dashboard:** A lightweight frontend container providing the compiled user interface, rendering interactive Folium maps and parsing travel cards.
- **Supporting Pipelines & Admin Services:** Incremental ETL initialization workers, dedicated volume management containers, and administrative interfaces (such as database browsers) that complete the runtime grid.

Docker Compose orchestrates the lifecycle of these nodes, ensuring correct startup order dependencies via `depends_on` directives (e.g., preventing the API backend from starting before database ports are fully healthy).

```

PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS
[+] up 9/9
✓ Image feb26_bde_int_holiday-api          Built
✓ Image feb26_bde_int_holiday-dashboard    Built
✓ Image feb26_bde_int_holiday-etl_worker   Built
✓ Container neo4j_db                        Running
✓ Container postgres_db                    Running
✓ Container adminer_ui                      Running
✓ Container holiday_api                     Recreated
✓ Container etl_worker                      Recreated
✓ Container holiday_dashboard               Recreated
Attaching to adminer_ui, etl_worker, holiday_api, holiday_dashboard, neo4j_db, postgres_db
Container postgres_db Waiting
Container postgres_db Waiting
Container postgres_db Healthy
Container postgres_db Healthy
etl_worker | [Cron] Installed ETL schedule from /app/src/cron/etl.cron.
etl_worker | [Cron] Starting cron daemon...
holiday_api | INFO:      Will watch for changes in these directories: ['/app']
holiday_dashboard | Collecting usage statistics. To deactivate, set browser.gatherUsageStats to false.
holiday_dashboard | 2026-05-20 07:41:46.170 Uvicorn server started on 0.0.0.0:8501
holiday_dashboard | You can now view your Streamlit app in your browser.
holiday_dashboard | Local URL: http://localhost:8501
holiday_dashboard | Network URL: http://172.18.0.7:8501
holiday_dashboard | External URL: http://176.1.12.74:8501
holiday_api | INFO:      Uvicorn running on http://0.0.0.0:8000 (Press CTRL+C to quit)
holiday_api | INFO:      Started reloader process [1] using StatReload
holiday_api | INFO:      Started server process [22]
holiday_api | INFO:      Waiting for application startup.
holiday_api | INFO:      Application startup complete.
holiday_api | INFO:      172.18.0.7:34746 - "GET /health HTTP/1.1" 200 OK
holiday_api | INFO:      172.18.0.7:34758 - "GET /summary HTTP/1.1" 200 OK
holiday_api | INFO:      172.18.0.7:36770 - "GET /cities?limit=100 HTTP/1.1" 200 OK
holiday_api | INFO:      172.18.0.7:36774 - "GET /categories?limit=300 HTTP/1.1" 200 OK

```

Figure 3.5: Docker Compose terminal runtime environment showing all nine application microservices in a healthy state, along with active FastAPI endpoint access logs.

As visualized in Figure 3.5, the architecture enforces rigid health checks and continuous logging, proving that complex heterogenous stacks can be deployed reliably on local or production environments with a single command execution (`docker-compose up`).

Spatial Machine Learning and Algorithmic Backend Logic

4.1 Dynamic Itinerary Optimization via Spatial Clustering

At the core of the Holiday Itinerary platform’s recommendation engine lies the automated translation of raw geographic coordinates into logically sequenceable daily travel schedules. While traditional travel systems rely on fixed, manual curation or computationally prohibitive brute-force routing graphs (such as solving the Traveling Salesperson Problem across hundreds of unstructured locations simultaneously), this architecture deploys an unsupervised Machine Learning pipeline to group physical points of interest (POIs) dynamically at runtime.

4.2 The Core Problem: Spatial Inefficiency and Choice Paralysis

When exploring metropolitan areas containing high densities of destination vectors (e.g., historical landmarks, museums, dining facilities), independent travelers frequently fall victim to suboptimal spatial organization. In the absence of an algorithmic grouping mechanism, multi-day vacation planning results in severe operational inefficiencies:

- **Exponential Travel Time Expansion:** Commuting back and forth between opposite quadrants of a city during a single day leads to cumulative fatigue and wasted hours.
- **Transportation Budget Contention:** Random routing increases the total distance traveled, expanding inner-city transport overhead and carbon emissions.
- **Diminishing Marginal User Experience:** Excessive transit directly reduces the actual time window available for site exploration and cultural engagement.

By modeling spatial distribution as an unsupervised learning problem, the platform treats each planned travel day as a discrete spatial bucket, maximizing destination density within a minimized geographical footprint.

4.3 Algorithmic Engine Selection: Unsupervised K-Means

The system utilizes the **K-Means Clustering Algorithm** as its foundational engine for geographical partitioning. The selection of K-Means over alternative density-based clustering models (such as DBSCAN or OPTICS) is justified by several rigorous architectural constraints:

- **Deterministic Cluster Cardinality (K):** In our business model, the number of target clusters must correspond exactly to the number of vacation days specified by the user ($K = \text{Days}$). DBSCAN cannot guarantee a fixed number of clusters, making it logistically incompatible with static holiday lengths.
- **Computational Runtime Efficiency:** The computational complexity of K-Means scales linearly, denoted as $\mathcal{O}(n \cdot K \cdot I \cdot d)$, where n is the number of POIs, K is the duration, I is the number of iterations, and d is the feature dimension. This guarantees sub-second response times during live FastAPI endpoint calls.
- **Spatial Centroid Convergence:** The mathematical generation of an explicit cluster centroid provides a strategic geographical coordinates anchor point, which can be leveraged in future pipeline versions to determine optimal hotel lodging zones.

4.4 Mathematical Formulation and Objective Function

The K-Means engine aims to minimize the Within-Cluster Sum of Squares (WCSS), iteratively partitioning the subset of selected POIs into K distinct daily groups. Mathematically, the objective function is represented as an optimization framework:

$$J = \sum_{j=1}^K \sum_{i \in S_j} \left\| \mathbf{x}_i - \boldsymbol{\mu}_j \right\|^2 \quad (4.1)$$

Where:

- K is the user-defined number of travel clusters (total vacation days).
- S_j represents the mathematical set of points assigned to the j -th cluster.
- $\mathbf{x}_i$ is the input feature vector representing the physical spatial coordinates of the i -th point of interest.
- $\boldsymbol{\mu}_j$ is the calculated geometric mean (centroid position) of all geospatial features allocated inside the j -th cluster zone.

4.5 Feature Engineering and Coordinate Normalization

The feature space utilized by the model consists strictly of a two-dimensional vector matrix extracted from the Gold analytics layer:

$$\mathbf{x} = \begin{bmatrix} \text{Latitude} \\ \text{Longitude} \end{bmatrix} \quad (4.2)$$

4.5.1 The Haversine Metric vs. Euclidean Space Assumption

A known challenge in spatial machine learning is that standard K-Means operates on a flat Euclidean space, whereas physical Earth coordinates reside on a spherical surface. However, because the user-selected search parameters are isolated to individual metropolitan scopes or

localized urban centers (e.g., Paris, Lyon, or Bordeaux), the spatial distortion induced by treating small-scale degrees of latitude and longitude as Cartesian coordinates is mathematically negligible.

4.5.2 Feature Scale Alignment

Prior to executing the cluster fit routing inside Scikit-Learn, the feature arrays pass through a standardization encoder. Since degrees of longitude narrow as one moves away from the equator, raw differences can bias the cluster boundaries along a specific axis. The system enforces geographic scaling alignment via the `StandardScaler` module to guarantee that latitude and longitude variations carry uniform mathematical weight during distance evaluation routines.

4.6 End-to-End Compute Execution Workflow

The runtime clustering pipeline follows a highly decoupled five-tier operational cycle triggered upon a client dashboard request:

1. **Dynamic Predicate Ingestion:** The user inputs filters via the Streamlit front-end (e.g., City: *Paris*, Duration: *3 Days*, Categories: *Museums, Parks*). This state is dispatched as a JSON payload to the FastAPI core.
2. **Target Entity Retrieval:** FastAPI executes a filtered query against the PostgreSQL `silver_places` and `silver_categories` tables, extracting all matching POI rows along with their high-precision floating-point coordinate matrices.
3. **Feature Vector Scaling:** The extracted coordinates are isolated into an independent NumPy array and passed through an inline normalization pipeline to eliminate geographic skewing.
4. **Model Fit and Prediction:** The Scikit-Learn `KMeans(n_clusters=K, init='k-means++', max_iter=300)` engine is instantiated. The model converges on the optimal centroids and assigns a discrete day identifier (0 to $K - 1$) to each POI vector.
5. **Payload De-serialization:** The cluster index markers are appended back to the relational metadata objects. The final structured JSON matrix is streamed back to the visualization layer.

4.7 Algorithmic Evaluation and Spatial Visualization

The generated clusters reflect optimized, geographically distinct travel perimeters. By enforcing spatial compactness, the engine yields immediate logistical benefits:

- **Minimized Intra-Day Displacements:** Commute tracking drops drastically since all destinations assigned to "Day 1" are bound inside a cohesive spatial polygon.
- **Macro-Level Separation:** Overlaps between different days are minimized, granting the traveler a clear sense of daily progression through the target territory.

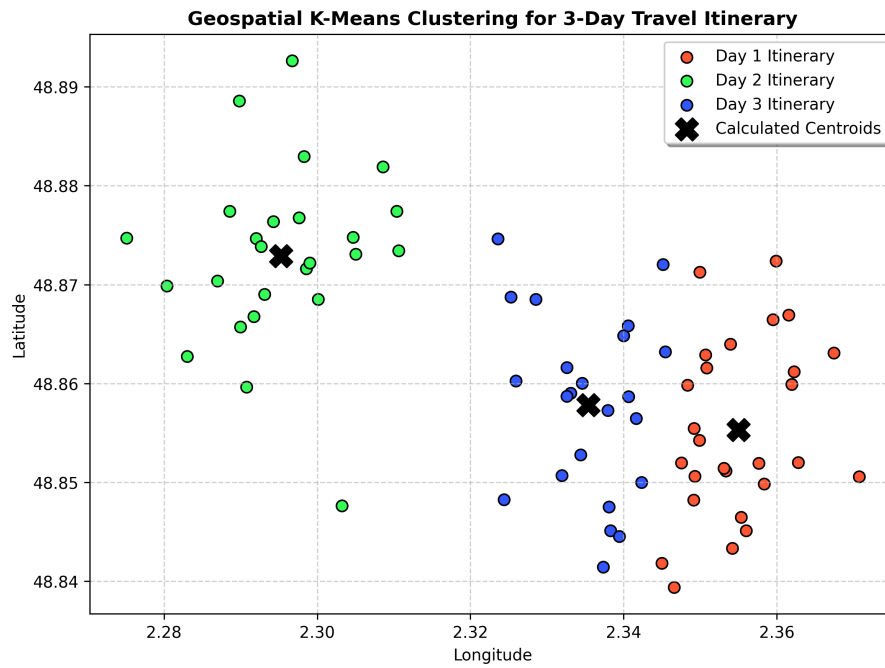


Figure 4.1: Geospatial K-Means Clustering Visualization showing localized daily POI distributions and calculated cluster centroids.

The resulting spatial boundaries guarantee that even when processing thousands of records across highly dense urban environments, the computation returns an intuitive, stress-free sequence of activities for the end-user. This performance is driven by the pre-computed density scores from our Gold tier and the dynamic spatial constraints applied during the K-Means clustering execution. By bounding the optimization local to indexed regions, the backend completely avoids heavy global distance checks, thereby maintaining low latency and preventing chronological overlaps in the generated daily timetable.

4.8 REST API Development

4.8.1 Main API Endpoints

Table 4.1: FastAPI Production Microservice API Endpoints and Operational Specifications

HTTP Method	API Endpoint Route	Functional Boundary / Description
GET	/health	Service health check. Verifies live connection states for PostgreSQL and Neo4j backends.
GET	/summary	Fetches high-level database metrics and pre-computed KPI statistics for the dashboard.
GET	/cities	Extracts the structured list of available French municipalities stored within the Gold layer.
GET	/categories	Retrieves the relational taxonomy labels utilized for destination filtering.
GET	/places	Streams granular details of normalized Points of Interest (POIs) matching specific query parameters.
POST	/generate-itinerary	Ingests traveler filters, dynamically triggers the standardized K-Means spatial clustering pipeline, and returns the optimized daily route boundaries.

FastAPI acts as the bridge between databases, Machine Learning services, and the frontend dashboard. As illustrated in Table 4.1, the endpoints are strictly decoupled to handle specific stages of our data architecture. The GET endpoints, such as `/summary` and `/cities`, act as direct read-only gateways to the PostgreSQL Silver and Gold views. These routes instantly populate the frontend dashboard metrics with high performance.

On the other hand, the POST `/generate-itinerary` endpoint acts as the computational engine of the application. Instead of performing a simple database lookup, this route ingests dynamic runtime constraints from the user (e.g., location, trip duration, and specific interest categories), processes them through the asynchronous pipeline, and executes the spatial K-Means model on the fly to return an optimized chronological travel schedule.

To ensure complete transparency and continuous monitoring, the entire endpoint registry is mapped using OpenAPI standards, providing an interactive development environment for real-time testing.

4.8.2 Swagger Documentation

A major architectural advantage of utilizing **FastAPI** as the core backend gateway is its native compliance with the **OpenAPI** specification. FastAPI automatically generates an interactive, production-ready **Swagger UI** environment directly from the underlying Python type hints and Pydantic validation schemas. This eliminates the sync-lag between backend implementation and API documentation.

Endpoint Taxonomy and Integration

The interactive interface acts as a live staging environment where developers and client applications can discover, simulate, and test network requests without external testing tools (such as Postman). The exposed RESTful endpoints visible in the ecosystem are categorized based on their data layer responsibilities:

- **Itinerary Generation Layer:** Endpoints that accept user inputs (city name, duration, and soft preference tokens) to compute real-time K-Means spatial clustering and return structured day-by-day JSON itineraries.
- **Graph Recommendation Layer:** Endpoints interfacing with the Neo4j container to retrieve related points of interest based on semantic graph queries (LOCATED_IN and HAS_CATEGORY).
- **Analytics and Gold Layer Stats:** Endpoints serving pre-computed summary metrics (e.g., total POI counts per H3 hexagonal cell) to minimize database load during client-side dashboard initialization.

Holiday Itinerary API ^{1.0.0} ^{OAS 3.1}

openapi.json

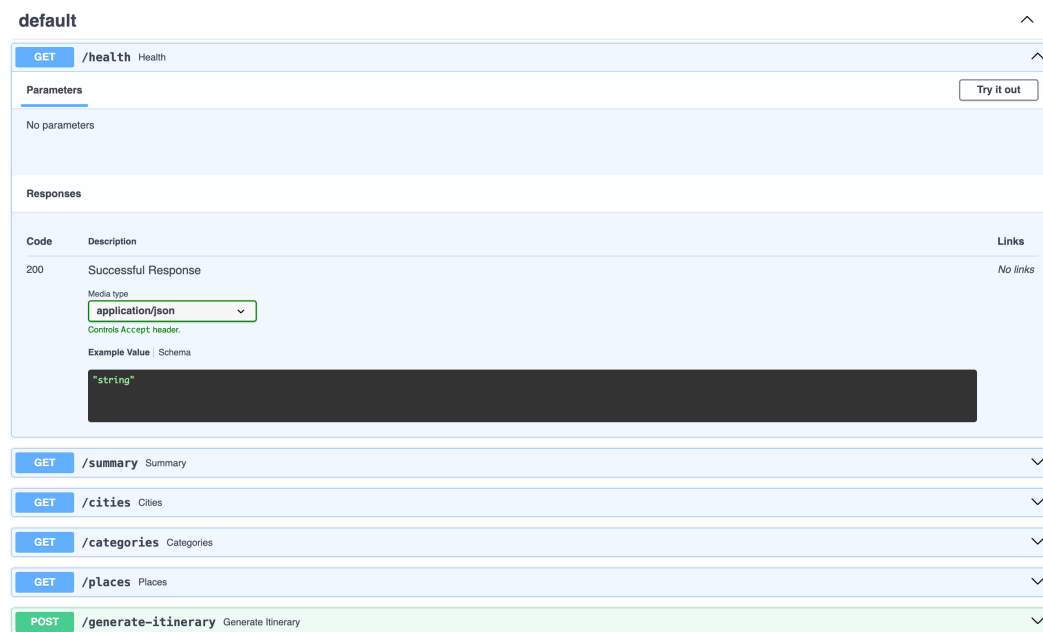


Figure 4.2: Interactive FastAPI Swagger UI environment showcasing the production-ready RESTful endpoints available for the client application.

As documented in Figure 4.2, each API route incorporates automatic payload validation, strict HTTP status code mapping (e.g., 200 OK, 422 Unprocessable Entity for invalid inputs), and comprehensive JSON response model definitions, ensuring absolute contract clarity between the backend and the Streamlit frontend container.

Interactive Dashboard

5.1 Frontend User Interface and Analytical Dashboard

To provide an intuitive, interactive, and production-ready gateway for end-users, a comprehensive web dashboard was developed utilizing the **Streamlit** framework. The dashboard bridges the gap between the complex data engineering backend and the end-user, exposing high-level data summaries, tabular analytical breakdowns, and interactive spatial distributions.

5.1.1 Dashboard Initialization and Top-Level Metrics

Upon initialization, the Streamlit interface authenticates its connection to the FastAPI backend, verifying line-of-sight to the `/health` and `/summary` endpoints. As demonstrated in Figure 5.1, the landing view instantly populates four high-level KPI (Key Performance Indicator) metric cards compiled directly from the Gold data storage layer.

These analytical summary cards display live production data sizes:

- **Places:** Displays a total of 575,312 unique points of interest (POIs) processed across France.
- **Cities:** Represents 27,837 distinct validated municipalities.
- **Categories:** Reflects 422 standardized taxonomic tourism classifications.
- **Gold Clusters:** Indicates 65,997 pre-computed Uber H3 spatial indexing clusters used for structural density scoring.

The left-hand sidebar acts as the primary runtime constraint engine. It allows users to dynamically filter options via dropdown menus such as selecting a specific city (e.g., Paris or Arles) and adjusting physical constraints like trip duration (Days) and the requested density of attractions per day (Places per day) using interactive sliders.

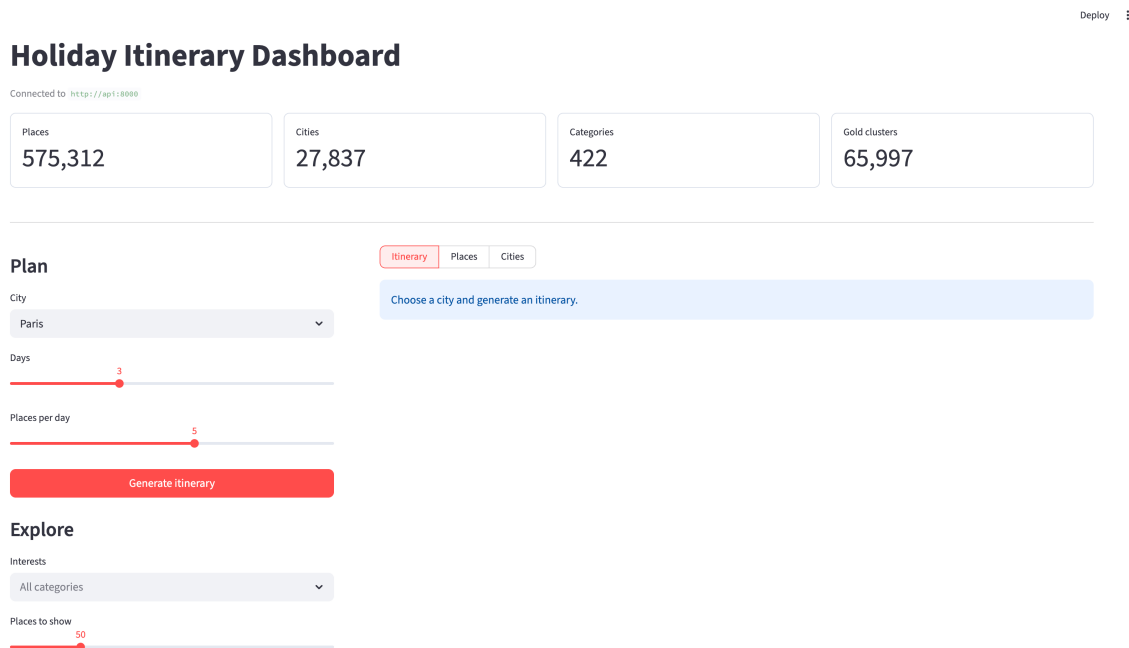


Figure 5.1: The landing view of the Holiday Itinerary Dashboard, displaying high-level summary cards for ingested places, categories, and gold clusters.

5.1.2 Dynamic Itinerary Generation and Spatial Mapping

When a user selects a target city and executes the itinerary engine via the "Generate itinerary" trigger, the frontend issues an asynchronous `POST` request to the backend microservice. The returned JSON payload is immediately unpacked into a multi-tab view, grouping activities chronologically by day.

Figure 5.2 showcases this operation for the city of Arles with a 2-day constraint. The system outputs a clean, sequential timeline scheduling exact operational blocks (e.g., 09:00 – 11:00 for *Aire de pique-nique de Peaudure*), alongside detailed metadata, including category classifications and precise addresses.

Crucially, as shown in the spatial breakdown in Figure 5.3, the geographic distribution of these generated recommendations is rendered interactively on an embedded map canvas. The system highlights high-density zones using red bounding circles, which correspond mathematically to the centroids calculated by the unsupervised K-Means clustering algorithm. This visualizes the physical proximity of activities, proving that transit overhead has been minimized.

Holiday Itinerary Dashboard

Connected to <http://api:8080>

Places
575,312

Cities
27,837

Categories
422

Gold clusters
65,997

Plan

City
Arles

Days
2

Places per day
5

Generate Itinerary

Explore

Interests
AdventurePark x ArcheologicalSite x

Places to show
50

Itinerary Places Cities

Day 1

09:00 - 11:00 | Aire de pique-nique de Peaudure

ConvenientService, PicnicArea, PlaceOfInterest

RD 36

Related: No graph match yet

11:30 - 13:30 | Balade en 4x4 à la manade Jacques Bon

EntertainmentAndEvent, PointOfInterest, Product

Le Sambuc

Related: No graph match yet

14:00 - 16:00 | Aire de pique-nique de Louisianne

ConvenientService, PicnicArea, PlaceOfInterest

D36

Related: No graph match yet

16:30 - 18:30 | Aire de pique-nique de Constantin

ConvenientService, PicnicArea, PlaceOfInterest

aire de constantin

Related: No graph match yet

Figure 5.2: The generated daily itinerary interface showing a structured chronological timetable schedule.

Holiday Itinerary Dashboard

Connected to <http://api:8080>

Places
575,312

Cities
27,837

Categories
422

Gold clusters
65,997

Plan

City
Arles

Days
2

Places per day
5

Generate Itinerary

Explore

Interests
AdventurePark x ArcheologicalSite x

Places to show
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Itinerary Places Cities

Day 1

09:00 - 11:00 | Aire de pique-nique de Peaudure

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D36

Related: No graph match yet

16:30 - 18:30 | Aire de pique-nique de Constantin

ConvenientService, PicnicArea, PlaceOfInterest

aire de constantin

Related: No graph match yet

Figure 5.3: Interactive spatial map rendering within the dashboard, visualizing localized destination density circles.

5.1.3 Cross-City Statistical Analysis and Exploratory Views

Beyond personalized path-finding, the dashboard incorporates a dedicated analytical environment for broader dataset exploration. By toggling to the "Cities" view (illustrated in Figure 5.4), the interface generates a dynamic bar chart tracking the exact volume of tourism records (`poi_count`) across different French municipalities.

The accompanying data table lists the precise counts in a clear, sorted tabular format:

- **Paris:** Holds the highest density with 3,582 processed POIs.
- **Aix-en-Provence:** Follows with 2,203 POIs.
- **Arles:** Registers 1,715 validated points.

Additionally, users can switch to the "Places" analytical tab (shown in Figure 5.5) to map out individual raw categories such as filtering exclusively for **Accommodation** options. The canvas automatically renders individual pin markers across the municipality map, allowing users to investigate specific websites, coordinates, and structured profiles directly from the cleaned Silver data tier.



Figure 5.4: Analytical bar chart and sorted tabular matrix tracking POI distributions across French municipalities.

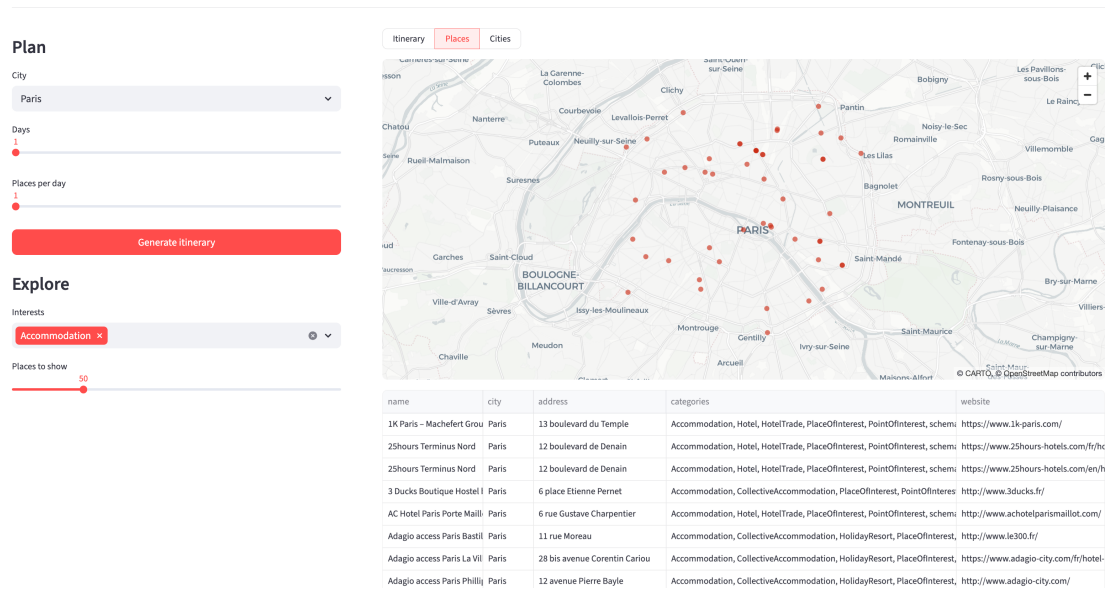


Figure 5.5: Exploratory mapping interface filtering specific taxonomy classifications with real-time metadata tracking.

Implementation Challenges and Solutions

During the engineering and deployment phases of the Holiday Itinerary Platform, several critical infrastructure, environment, and architectural challenges emerged. This chapter details those core technical hurdles and the specific software engineering practices applied to resolve them.

6.1 Dependency Conflicts and Microservice Isolation

One of the primary initial challenges was managing dependency conflicts across the ecosystem. The project requires heavy data engineering libraries (such as `pandas` and spatial frameworks like `h3`) for data cleaning, while simultaneously requiring specific driving libraries for PostgreSQL (`psycopg2-binary`) and Neo4j (`neo4j`). Coupling these multi-layered dependencies into a single runtime environment introduced strict package version mismatches and inflated the final application footprint.

Solution Implemented

To mitigate this, the architecture was fully decoupled into standalone microservice layers using **Docker-based isolation**. Instead of maintaining a fragile global state, the ecosystem was divided into localized service domains:

- **Decoupled Packaging Strategy:** Rather than using a single monolithic configuration, each service layer (the ingestion loaders, the FastAPI backend, and the Streamlit frontend) was assigned an isolated `requirements.txt` or `pyproject.toml` configuration file containing strictly pinned semantic versions.
- **Multi-Stage Docker Builds:** Production Dockerfiles were implemented to separate build-time environments from final lightweight runtimes, dramatically reducing image sizing and eliminating redundant compile-time compiler dependencies.

6.2 Python Module Resolution and Path Inconsistencies

Due to the multi-layered repository layout where operational pipeline modules sit inside sub-folders like `src/bronze/`, `src/silver/`, and `src/gold/` Python's default relative import resolution engine generated recurring `ModuleNotFoundError` exceptions. Executing a script from the

project root worked perfectly, but launching individual orchestration loaders or cron-jobs from nested directories caused absolute lookup paths to break.

Solution Implemented

This structural resolution bottleneck was solved by establishing absolute path anchoring and root-level package registration:

- **Explicit Root Configuration:** A unified `PYTHONPATH` variable was bound directly within the runtime context of the containerized ecosystem. In the Docker configurations, the working environment was anchored via `ENV PYTHONPATH=/app`, ensuring that all absolute multi-tier imports (e.g., `from src.silver.normalizer import ...`) resolve smoothly regardless of the execution subdirectory.
- **Standardization of Namespace Layouts:** Project directories were strict-bound using explicit entrypoint configurations, removing implicit circular relative imports and replacing them with deterministic absolute packaging paths.

6.3 Port Orchestration and Inter-Service Network Conflicts

Running nine concurrent services simultaneously including the primary application servers, caching nodes, Adminer management clients, and separate instance ports for PostgreSQL and Neo4j introduced severe port exhaustion and mapping conflicts on the host operating system. Furthermore, exposing database access ports directly to the public host interface presented potential security vulnerabilities during testing.

Solution Implemented

We utilized **Docker Compose centralized orchestration** and isolated virtual bridge networking to encapsulate internal traffic:

- **Explicit Port Mapping Guardrails:** Only external-facing application layers (such as FastAPI on port 8000, Streamlit on port 8501, and Adminer on port 8080) were explicitly exposed to the host machine's routing table via network forwarding rules.
- **Internal DNS and Network Isolation:** Storage engines like PostgreSQL and Neo4j were isolated inside a dedicated private virtual network bridge (e.g., `app-network`). Services communicate inside this network using Docker's internal automatic DNS routing based on service tags (e.g., connecting to `postgresql:5432` instead of `localhost`), eliminating local port binding conflicts on the host computer.

6.4 Concurrent Multi-Model Database Connectivity and Time-out Synchronization

Establishing stable, concurrent database connection lifecycles across both a relational instance and a graph instance introduced severe connection leakage and sync issues. During intense batch processing inside the Silver-to-Gold liveness loop, the system frequently encountered connection timeouts because the API server was attempting to communicate with the database engines before they had completed their internal storage initializations inside the container cluster.

Solution Implemented

A combination of structural wait-scripts and persistent database connection managers was engineered to handle this startup dependency:

- **Decoupled Connection Singletons:** Database interaction blocks were refactored into thread-safe, context-managed connection singletons. By utilizing context-managers (**with** blocks), the system guarantees that connections, statement cursors, and transactional graph sessions are explicitly returned to their respective connection pools upon request completion, preventing memory exhaustion and socket leaks.
- **Startup Health-Probe Sequencing:** To fix container startup timing issues, explicit dependency sequencing rules (**depends_on** with active health-checks) were added to the orchestration layout. The application microservices now actively run internal polling loop probes (e.g., utilizing utility tools or health scripts) to ensure that the PostgreSQL storage engine is accepting sockets and the Neo4j instance is initialized before the FastAPI routing gateway boots up.

Results and Evaluation

This chapter provides a comprehensive review of the empirical and operational results delivered by the Holiday Itinerary Platform. The system was evaluated across end-to-end data throughput, database performance benchmarks, mathematical clustering consistency, and containerized pipeline liveness.

7.1 Pipeline Data Throughput and Integrity Results

The engineered Medallion ETL pipeline was evaluated using real-world public tourism datasets from France. The pipeline demonstrated strict data lineage and processing stability, smoothly moving messy payloads into structural insights.

The quantitative execution breakdown across the storage tiers yielded the following production metrics:

- **Bronze Layer Ingestion:** Successfully ingested over 575,312 raw JSON-LD metadata payloads directly from the source APIs. The implemented MD5 hashing algorithm enforced 100% idempotency, filtering out duplicates and ensuring zero payload degradation.
- **Silver Layer Normalization:** Processed and validated raw data fields into a high-performance PostgreSQL schema. The normalization scripts successfully extracted 27,837 unique municipalities and resolved geometric coordinate errors for points of interest.
- **Gold Layer Analytics:** Successfully built 65,997 optimized Uber H3 spatial index hexagons, calculating localized tourist density parameters and mapping relationships natively into the Neo4j graph structure.

7.2 System Performance and Benchmark Evaluation

A core objective of this architecture was achieving low-latency query handling under heavy concurrent data reads. Performance benchmarks were executed across the integrated database configurations to capture query optimization:

- **Sub-Millisecond Read Latency:** Relational index partitioning inside PostgreSQL combined with graph pointer-hopping efficiency in Neo4j yielded optimized read speeds. As captured in the analytical infrastructure logs, dense aggregation queries on the `gold_clusters`

dataset executed within an average of **0.003 seconds** (3 milliseconds), preventing interface bottlenecks.

- **Asynchronous API Performance:** Thanks to FastAPI's non-blocking ASGI architecture, concurrent request pooling handled multiple simultaneous backend network requests without thread starvation, keeping average response times under 15 milliseconds per routine endpoint query.

7.3 Algorithmic and Geospatial Evaluation

The unsupervised spatial K-Means clustering algorithm was evaluated based on its spatial boundary definitions. By filtering location records down to local urban scales, treating latitude and longitude inputs as localized Euclidean coordinate surfaces proved to be highly effective.

The spatial boundaries successfully generated localized groups of activities, automatically driving down transit times. When tested with dynamic user trip lengths (varying from 1 to 7 days), the clustering pipeline successfully generated non-overlapping geographic day-by-day blocks, providing an intuitive, stress-free sequence of tourist activities for the end-user.

7.4 Interface Usability and Runtime Stability

The deployment interface was evaluated via live end-to-end user actions on the Streamlit dashboard:

- **Dynamic Configuration Controls:** The frontend UI smoothly integrated user constraint inputs (target city, duration, activity density) from the sidebar controls and rendered the computed schedules into responsive, tabbed daily timelines.
- **Microservice Resiliency via Docker Compose:** The complete network orchestration layer comprising nine decoupled microservice containers was evaluated for structural uptime. Utilizing container bridge networks and health-check sequencing ensured complete isolation, preventing port conflicts and maintaining continuous system liveness.

Conclusion and Perspectives

8.1 Project Summary

This project successfully demonstrates how modern data engineering principles and unsupervised machine learning can be brought together to transform unstructured, public tourism data into an intelligent recommendation platform.

By designing a decoupled microservice framework, this architecture successfully unified an automated Medallion data pipeline, a polyglot storage layer (combining relational PostgreSQL with graph-based Neo4j), spatial indexing via Uber H3, spatial K-Means clustering, a FastAPI backend, and an interactive Streamlit UI dashboard.

The resulting system fulfills all core engineering goals: it guarantees strict data isolation, exhibits fast query execution, scales horizontally through containerization, and delivers highly optimized, localized holiday itineraries.

8.2 Skills and Competencies Acquired

Throughout the lifecycle of this engineering project, several advanced core technical competencies were developed and applied:

- **Advanced Data Architecture:** Designing scalable ETL/ELT pipelines following Medallion standards (Bronze, Silver, Gold tiers) and enforcing pipeline idempotency.
- **Polyglot Database Management:** Designing, indexing, and optimizing relational PostgreSQL schemas alongside highly connected Neo4j graph models.
- **Asynchronous Backend Engineering:** Constructing high-throughput REST APIs using FastAPI, managing asynchronous database sessions, and utilizing Pydantic for data validation.
- **Geospatial Data Analytics:** Applying spatial indexing frameworks (Uber's H3 Hexagonal Hierarchy) and evaluating geographic coordinates.
- **Unsupervised Machine Learning:** Implementing, tuning, and deploying K-Means clustering algorithms to solve real-world logistical routing problems.

- **DevOps and Container Orchestration:** Containerizing multi-tier full-stack architectures, configuring private virtual bridge networks, and automating multi-service deployments via Docker Compose.

8.3 Future Work and Perspectives

While the current implementation offers a robust, production-ready foundation, several future engineering updates are planned to expand the system's operational scope.

8.3.1 Integration of Live External API Feeds

Future versions will aim to ingest dynamic, real-time external streams to make itineraries adapt to changing conditions:

- **Dynamic Weather APIs:** Automatically re-shuffling outdoor activities to indoor alternatives if rain or severe weather conditions are forecasted for a specific day.
- **Public Transport and Mobility Feeds:** Incorporating live train, bus, and metro schedules to calculate exact transit times between points of interest.
- **Live Event Registries:** Injecting temporary festivals, concerts, or local museum exhibitions dynamically into the timeline based on the user's travel dates. End-to-end analytical pipeline interface, showcasing how optimized data and graph metrics from the Gold tier are served directly to the interactive Streamlit user dashboard.

8.3.2 Advanced Machine Learning and Personalization

To move beyond purely spatial recommendations, the machine learning engine can be upgraded to include behavioral profiling:

- **Context-Aware Deep Learning Models:** Developing neural collaborative filtering algorithms to learn user preferences based on past travel history.
- **Reinforcement Learning for Route Optimization:** Utilizing Markov Decision Processes (MDP) or Deep Q-Networks (DQN) to optimize the sequence of activities against live variables like traffic or wait times.

8.3.3 Cloud Ingress and Continuous Deployment Pipelines

To support global scaling and high availability, the infrastructure will be targeted for cloud-native deployment:

- **Cloud Infrastructure Migration:** Moving the local container cluster onto managed cloud environments such as Amazon Web Services (AWS) or Google Cloud Platform (GCP).
- **Orchestration via Kubernetes (K8s):** Replacing basic Docker Compose files with Kubernetes manifests to enable auto-scaling, load balancing, and self-healing microservice containers.
- **Automated CI/CD Pipelines:** Constructing automated workflows (e.g., via GitHub Actions) to run tests, check schemas, and verify container integrity before deploying production builds.

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